Subfield cover crops for biofuel: A three-part technical review

The proposed work — DSSAT-driven, SSURGO-informed delineation of within-field zones to optimize **spatially variable placement of biofuel cover crops** (cereal rye, winter camelina, pennycress) in Midwest corn-soybean rotations — sits in a genuine and publishable white space, but rests on a methodologically uneven foundation. The crop-modeling architecture is defensible for cereal rye and the cash crops; it is essentially aspirational for camelina and pennycress. The soil-input choice (SSURGO) captures landscape-scale, not true within-field, variability and must be supplemented. The strongest publishable contribution is conceptual: extending the Brandes/Bonner/Schulte "subfield diversification" framework from perennials (switchgrass, miscanthus) to **winter annual biofuel cover crops**, which preserve 100% of grain area and lower the adoption barrier for biofuel feedstock production. The sections below assess methodology, feasibility, and novelty in turn, with concrete recommendations to make the work both rigorous and novel.

Section 1 — Methodological validity and state-of-the-art currency

SSURGO is a defensible baseline but does not deliver true within-field variability

SSURGO and its raster repackagings (gSSURGO at 10 m, gNATSGO at 10 m with STATSGO2 gap-fill) **are vector soil-survey products rendered at 1:12,000–1:63,360 with minimum delineations of 1–4 acres**, and each map unit is a *complex of named components* (e.g., 60% Tama / 30% Muscatine / 10% Garwin) that are not spatially explicit within the polygon. The 10 m raster resolution of gSSURGO is illusory: the underlying polygons are unchanged. For DSSAT, this forces an aggregation choice — dominant component, area-weighted average, or component-wise simulation with area-weighted output — and each choice systematically masks catena-scale variation (eroded summit vs. foot-slope) that is exactly what variable-rate cover-crop placement should target. ScienceDirect

Peer-reviewed evaluations are consistent on this point. **Stermitz et al. (1999)** found SSURGO-based yield predictions did not improve precision over whole-field means on Montana dryland wheat. **Sudduth, Kitchen et al. (2003, 2005)** showed on-the-go ECa and elevation explained Missouri claypan yield variability that SSURGO did not capture, and SSURGO map-unit boundaries did not align with multi-year yield-derived productivity zones. **Pachepsky et al. (2017, *Geoderma*)** demonstrated that SSURGO curve number dominated DSSAT yield prediction, indicating soil hydraulic detail is being lost in upscaling. **A 2025 *Agronomy Journal* paper on DSM-based clustering found 78–236% reductions in within-zone yield variability versus SSURGO-based zones.** The honest conclusion: SSURGO/gNATSGO captures **between-field and landscape variation well; it captures genuine sub-field variation poorly**.
Wiley Online Library + 2

The recommended soil-data hierarchy is layered, not SSURGO-only

POLARIS (Chaney et al., 2016 *Geoderma*; 2019 *Water Resources Research*) is the natural complement — a **30 m probabilistic remapping of SSURGO** via DSMART-HPC random forest that delivers van Genuchten/Brooks-Corey hydraulic parameters with 5th/50th/95th percentile bounds, spatially disaggregating the SSURGO complex problem. POLARIS is constrained by the same SSURGO legacy data, but its 30 m grid and explicit uncertainty are precisely what the DSSAT-zone workflow needs. SoilGrids 2.0 (Poggio et al., 2021, *SOIL*) at 250 m is too coarse for CONUS field work when POLARIS exists. **Field-measured ECa from Veris, EM38, or DUALEM** — the canonical Corwin & Lesch (2005, *Comput. Electron. Agric.*) methodology — remains the gold standard for true within-field variation and should be used on a validation subset. **Topographic wetness index from 3 m LiDAR** consistently emerges as one of the strongest single predictors of Midwest within-field yield stability (Maestrini & Basso, 2018, *Sci. Reports*). Scholars@Duke + 4

Recommendation: Use POLARIS as the primary 30 m soil layer with gNATSGO for coverage, supplement with LiDAR-derived TWI, and validate on at least a few fields with ECa or yield-monitor data. Propagate POLARIS 5th/95th percentile bounds through DSSAT to quantify how much zone delineation depends on soil-input uncertainty — this is rarely done and would itself be a methodological contribution.

Daymet V4 is the best gridded weather choice, but field-scale soil variability dominates within-field weather variability

For Midwest spatial DSSAT, the definitive comparison is **Mourtzinis, Rattalino Edreira, Conley & Grassini (2017, *Eur. J. Agronomy* 82:163–172)**: across 45 station-weather datasets, Daymet and PRISM had small biases for temperature and growing-degree-days but **poor performance for relative humidity, precipitation, and reference ET (ETo bias of +253 mm with PRISM vs. −4 mm with Daymet)**. Simulated maize-yield RMSE was 18% of mean yield with Daymet vs. 24% with PRISM, with 20–32% of site-years showing >15% yield deviation. Their stark conclusion was that gridded weather data "cannot replace measured weather data as a basis for field-scale agricultural applications." NASA POWER at ~50 km is unsuitable (van Wart et al., 2013, *Field Crops Res.*); gridMET, NLDAS-2, and ERA5 are coarser and inherit PRISM precipitation. **Daymet V4 (1 km)** is the right gridded choice, but for a single ~50 ha field a single bias-corrected station-anchored weather file is defensible and avoids spurious 1 km grid-edge artifacts.

DSSAT is methodologically current for corn and soybean, weak for cover crops compared to alternatives

For corn-soybean rotations in the Midwest, **CERES-Maize and CROPGRO-Soybean within**

DSSAT-CSM v4.8.5 are mature and well-validated (Jones et al., 2003, *Eur. J. Agronomy*;

Chatterjee et al., 2025, *Agric. Water Mgmt.*; Birru et al., 2023, *Agronomy Journal*). The **CENTURY-based soil organic matter module** (Gijsman, Hoogenboom, Parton & Kerridge, 2002, *Agron. J.* 94:462–474), with three SOM pools and surface residue layer, is the appropriate engine for multi-year cover-crop rotations, validated against 40-year bare-fallow datasets at r=0.983 for SOM-C. SOC pool initialization following Basso & Gargiulo (2011) is mandatory because incorrect initial pool partitioning can dominate long-term simulations. Tillage in DSSAT (Porter et al., 2010; White et al., 2010) is implemented as residue mixing and bulk-density modification, but **post-tillage re-settling dynamics are simplified — a known weakness versus APSIM and RZWQM2**. Dssat + 5

The honest evaluation for cover crops is more critical. **DSSAT has no native cereal rye, winter camelina, or pennycress module.** All published DSSAT cereal rye work uses CERES-Wheat as a proxy with cold-tolerance and phenology adjustments (Gupta et al., 2023; Birru et al., 2023; Riedesel et al., 2024, *Agronomy Journal*, which explicitly states "rye crop modeling is still in its infancy"). **Kharel et al. (2022, *Agric. Water Mgmt.*) candidly noted: "to our knowledge, there are no studies that have used DSSAT, without its integration with any other models, to simulate cereal rye growth as a winter cover crop"** — a quotable gap statement. Birru et al. (2023) further found DSSAT could not replicate measured cereal-rye effects on subsequent cash-crop yields because residue-N mineralization was overestimated. DSSAT does not support simultaneous intercropping; only sequential rotations are simulated. For the Midwest cover-crop literature, **APSIM (Archontoulis group at Iowa State; Basche, Martinez-Feria, Kakarla, Plastina) and RZWQM2 (Malone, Kaspar, Qi at USDA-ARS Ames) have substantially deeper validation records**. Because DSSAT is fixed by the project specification, the methodological burden is to (a) make the cereal-rye CERES-Wheat parameterization explicit and rigorous, (b) build first-of-their-kind CROPGRO-canola-based parameterizations for camelina and pennycress, and (c) explicitly compare results to APSIM/RZWQM2 benchmarks rather than ignore them. Wiley Online Library + 2

Crop-model-driven management zones are an emerging but established methodology

Identifying local optima per soil unit and clustering responses into zones traces back fifteen years through **Booltink et al. (2001), Basso et al. (2001, 2007 *Eur. J. Agron.*, 2012)**, and **Thorp et al.'s Apollo DSS (2008, *Comput. Electron. Agric.* 64:276–285)**, which was explicitly designed to automate DSSAT spatial simulations across zones, calibrate to historic yield maps, and generate variable-rate prescriptions. Recent work in the same lineage includes **Guerrero et al. (2022, *Agric. Syst.*)** combining remote-sensing zones with auto-calibrated DSSAT for cotton N rate; **Jiang et al. (2024, *Precision Agriculture*)** using APSIM for site-specific corn EONR; and **Mandrini, Bullock & Martin (2021)** for variable-rate N in Midwest corn. **Fuzzy c-means clustering via the Management Zone Analyst software (Fridgen et al., 2004, *Agron. J.* 96:100–108)** remains the de-facto baseline, with recent reviews by **Nawar et al. (2017)** and **Moral et al. (2020, 2023)** documenting refinements. The proposed approach is methodologically defensible. The unresolved critique is that crop-model outputs are only as good as soil and weather inputs, so SSURGO/POLARIS uncertainty propagates directly into zone boundaries — **multi-year temporal stability analysis (Blackmore 2000; Maestrini & Basso, 2021, *Precision Agriculture* 22:1749) is essential to identify zones robust across weather variability rather than artifacts of a single growing season.** USDA ARS + 4

Section 2 — Implementation feasibility

The SSURGO-to-DSSAT pipeline is buildable but requires custom scripting

No turnkey tool ingests SSURGO polygons and emits DSSAT .SOL profiles. The **mature open-source workflow uses the R packages `soilDB` (Beaudette, Skovlin, Roecker) and `aqp` (Algorithms for Quantitative Pedology)** to query USDA-NRCS Soil Data Access, retrieve horizon-level data by mukey/cokey, perform depth-weighted aggregation across components, and feed a custom .SOL writer. A Python alternative is `DSSATTools` (Py_DSSATTools on GitHub) with the Alderman `DSSAT` R package on CRAN providing equivalent functionality. **Bespoke published pipelines include Sheshukov et al. (2011) and Han et al. (2019)** for global SoilGrids-to-DSSAT conversion. **The Soil Profile Optimizer plug-in to DSSAT v4.8 (Precision Agriculture, 2023)** addresses a key well-known issue — raw SSURGO-derived soil inputs predict yield poorly (R^2=0.03) until hydraulic parameters are calibrated against observed yield/biomass per zone (R^2=0.76 after SPO optimization). **Thorp & Bronson's GeoSim (2013, *Environ. Modeling & Software*)** and **Thorp et al.'s Apollo (2008)** demonstrate the polygon-based spatial DSSAT pattern at field scale; both remain useful templates. The recommended pipeline is **soilDB + aqp → depth-weighted .SOL writer → SPO calibration → DSSAT FileX via DSSATTools or XBuild.** Springer ScienceDirect

Computational feasibility is well-solved by the parallel-gridded DSSAT framework

For spatial DSSAT across many soil × management × weather combinations, two frameworks are mature. **pSIMS** (Elliott et al., 2014, *Environ. Modeling & Software*) was developed at UChicago RDCEP/Argonne for global gridded crop modeling (GGCMI, ISIMIP) and bundles AgMIP translators; it is purpose-built for regional grids and somewhat overkill for sub-hectare cells. **Alderman's parallel gridded DSSAT-CSM (2021, *Geoscientific Model Development* 14:6541–6569)** uses MPI + NetCDF, achieves 97–99% parallel efficiency, and is roughly 13× faster than serial on 16 cores. For thousands-to-millions of combinations across a watershed, Alderman's framework is the right choice. A typical field-scale experiment (10 fields × 20 zones × 30 years × 12 management combinations = 72,000 DSSAT runs) is well within a single workstation's capability over hours to days; scaling to county or state is feasible on shared-memory HPC. Copernicus Copernicus

Cover-crop calibration is the biggest feasibility obstacle, especially for the oilseeds

For cereal rye, the calibration pathway exists: **CSM-CERES-Wheat with adjusted P1V, P1D, P5, and cold-tolerance parameters** (Gupta et al., 2023; Birru et al., 2023; Riedesel et al., 2024). Field validation data are abundant — **USDA-ARS Ames (Kaspar, Malone, Jaynes); Iowa State Marsden Farm; UNL Nebraska (Blanco-Canqui, Ruis); Illinois State Lexington (Armstrong, Ruffatti); Univ. of Minnesota (Yu, 2024); Practical Farmers of Iowa on-farm trials**. Typical biomass at termination ranges 0.4–4.5 Mg ha^{-1} depending on planting date and "planting green" practice. ScienceDirect + 3

For **winter camelina and pennycress, the obstacle is fundamental**: no published DSSAT calibration exists for either crop. CROPGRO-canola (Brassica napus) has been calibrated (Deligios et al., 2013, *Field Crops Res.*; Saseendran et al., 2010, *Agron. J.*) and is the logical adaptation framework. Boote et al.'s (2021) SPARC work on CROPGRO for B. carinata is the closest published precedent and stated bluntly: "at present, there are no crop models for carinata." Pennycress modeling is essentially nonexistent across all crop models — only physiological/molecular work (McGinn et al., 2018, *Plant Biotechnol. J.*; Dorn et al., 2015, *DNA Res.*) and empirical GDD relationships exist (Phippen et al., 2022, *Frontiers in Energy Research* 10:793776). Field datasets for calibration are growing rapidly through USDA-ARS Morris (Gesch, Forcella, Eberle, Weyers), the Forever Green Initiative, IPREFER (the $10M USDA-NIFA project led by Phippen at WIU), and Berti's NDSU work — typical camelina seed yields are 900–1,700 kg/ha with 2–4 Mg/ha total biomass; pennycress yields 800–2,200 kg/ha. **Building first-of-their-kind CROPGRO-Camelina and CROPGRO-Pennycress cultivar coefficients is itself a publishable methodological contribution**, but it transforms the project from an applied modeling study into a substantially larger crop-model development effort. Wiley Online Library + 7

Biomass harvest is modelable but cover-crop N mineralization remains a known DSSAT weakness

DSSAT can mechanically simulate cover-crop biomass harvest via the "harvest residue left in field" parameter, partitioning aboveground biomass between removal and surface residue. The CENTURY module then propagates resulting C and N dynamics into subsequent cash-crop simulation. **CROPGRO-Perennial Forage (Lara, Pedreira, Boote et al.)** supports repeated harvests with regrowth. The critical caveat is that **Birru et al. (2023) found DSSAT over-mineralizes cereal-rye residue N**, so harvest-vs.-residue contrasts may be exaggerated — explicit comparison to measured paired-treatment data is essential. The cleanest North-Central US precedent for harvested-rye energy cover crops is **Malone et al. (2018, *Agric. Environ. Lett.*; 2023, *Environ. Res. Lett.*)** using RZWQM2 with embedded CSM-CERES plant models across 41 Midwest sites; their finding that fertilized harvested rye reduced N leaching 54% versus bare fallow and 18% versus unharvested unfertilized rye is directly transferable framing. For oilseed CCs, only ~10–15% of total biomass (the seed) is harvested, minimizing SOC penalty. Dssat + 4

Variable-rate cover-crop seeding is technically feasible but economically marginal

Commercial equipment exists: **Montag Manufacturing Gen 2 dry delivery with map-based

variable-rate metering; Hagie-Montag interseeders (e.g., the Polk County, Iowa community machine deployed in 2022); Hiniker 3-point toolbar interseeders; John Deere ExactEmerge with SeedStar 5; Great Plains drills with ISOBUS prescriptions; Kinze electric-drive planters; Precision Planting RateController retrofits. Drone seeders (DJI Agras T40) are increasingly used for in-season interseeding. **Mark Licht's Iowa State trials (Integrated Crop Management)** evaluated four prescription methods (uniform, normalized production history, Corn Suitability Rating, TWI) and found similar yields across methods, with topographic wetness giving the smallest standard error and highest return to seed. Integrated Crop Management

The economic constraint is real. **Cover crop seed costs are only $20–35/acre in Iowa**, so the dollar value of varying the rate within a field is small unless the prescription drives a major decision such as zeroing out cover crops in low-response zones or concentrating biofuel-feedstock CCs in high-biomass zones. **Plastina (ISU CARD)** concluded cost-share payments are insufficient to cover private costs, and Iowa adoption was still only ~1.55% of row-crop farmland as of the 2016 NASS (substantially higher now but far from saturation). Roughly half of Iowa farmland is rented, adding tenant-landlord coordination as a structural barrier. **The economic case is much stronger when CCs deliver an additional revenue stream** — CoverCress pennycress contracts via Bayer/Bunge/Chevron, SAF 45Z tax credits, climate-smart commodity payments — which is the core of the proposed project's value proposition.
Iowaagricultureauthority + 5

Operational match between modeled prescriptions and harvest windows

The most underappreciated feasibility constraint is the **harvest window**. Cereal rye must be terminated by early-to-mid May for corn, with documented yield drags from later termination. Winter camelina and pennycress mature in late June, conflicting with soybean planting — both crops require **relay cropping with soybeans interseeded in April–May**, reducing soybean yield to 58–83% of monocrop but increasing total system oil yield ~50% (Gesch & Archer, 2013, *Ind. Crops Prod.*; Ott et al., 2019, *Agron. J.*; Mohammed et al., 2022, *Food and Energy Security*). The proposed modeling must explicitly handle these temporal constraints in the DSSAT FileX management sequences and reflect them in the prescription logic. AgUpdate + 4

Section 3 — Novelty and publishability gap analysis

The exact intersection is genuinely unoccupied

The proposed work sits at a four-way crossroads: (i) process-based crop modeling with high-resolution soils, (ii) subfield management-zone delineation, (iii) spatially variable placement of **winter annual** cover crops, and (iv) bioenergy/biofuel feedstock optimization. **No published study combines all four.** Each individual component is well-represented in the literature, which

is actually a strategic strength — the project occupies the white space between three mature is actually a strategic strength — the project occupies the white space between three mature streams.

Closest analog: the Iowa subfield-diversification trilogy (Brandes, Bonner, Schulte, Muth, Heaton)

The most important antecedent is **Brandes, McNunn, Schulte, Bonner, Muth, Babcock, Sharma & Heaton (2016), "Subfield profitability analysis reveals an economic case for cropland diversification," *Environmental Research Letters* 11:014009**. Using AgSolver's Landscape Environmental Assessment Framework (LEAF), they mapped ~3.3 million subfield polygons across 9.3 Mha of Iowa cropland and showed that **substantial subfield acreage in corn-soy systems is unprofitable at recent prices** and that aggregate field profit could increase by diversifying these zones. Their follow-up — **Brandes et al. (2018), *GCB Bioenergy* 10:199–212** — used LEAF + DNDC to evaluate **targeted subfield switchgrass integration**, finding that converting 37% of cropland to switchgrass could yield 34 million Mg biomass and a 38% statewide NO_3-N reduction. The **Bonner et al. (2014, *Energies* 7:6509; 2016, *J. Soil and Water Conservation* 71:182) and Griffel et al. (2017, *BioEnergy Research*)** papers extend the LEAF framework to integrated bioenergy production systems with switchgrass, miscanthus, sorghum, and willow on identified subfield zones. The broader **STRIPS prairie-strips work (Schulte et al., 2017, *PNAS* 114:11247)** is the strategic within-field perennial-placement analogue, demonstrating that 10% of cropland in prairie strips delivers 95% sediment reduction, 90% phosphorus reduction, and 85% N runoff reduction with no significant yield loss. Illinois Experts + 5

The Brandes/Bonner/Schulte lineage uses LEAF and DNDC, not DSSAT, and exclusively targets perennial bioenergy crops. Switching the conservation/bioenergy target from perennials to **winter annual biofuel cover crops** is not a trivial extension — it preserves grain production on 100% of acreage, requires no multi-year land-use commitment, and lowers the adoption barrier dramatically compared to a 37% perennial conversion. This is the strongest single argument for novelty.

Closest analog from DSSAT precision-ag lineage

The Basso/Thorp tradition — **Thorp et al. (2008) Apollo DSS, Basso et al. (2007, 2012, 2016, 2019), Maestrini & Basso (2018, 2021), Guerrero et al. (2022, *Agric. Syst.*)** — has applied DSSAT and SALUS to within-field zones for **cash-crop variable-rate N, seeding, and irrigation**. None addresses cover crops as the management decision variable. The 2025 *Agronomy Journal* DSM-clustering paper showed digital soil maps outperform SSURGO for zone delineation — a methodological lesson the proposal should adopt. ScienceDirect

Closest analog from biofuel cover-crop lineage

The Forever Green Initiative (UMN, Wyse, Hunter, Anderson, Wells); USDA-ARS Morris (Gesch, Forcella, Eberle, Weyers); NDSU (Berti); WIU (Phippen, IPREFER); CoverCress

Inc./Bayer/Bunge/Chevron have built the agronomic, breeding, and supply-chain case for camelina and pennycress as biofuel feedstocks. Key publications include Gesch & Archer (2013, *Ind. Crops Prod.* 44:718); **Sindelar et al. (2017, *GCB Bioenergy* 9:508)** on winter oilseed production for biofuel in the Corn Belt; Johnson et al. (2017, *Agron. J.*); Ott et al. (2019, *Agron. J.* 111:1281); **Phippen et al. (2022, *Frontiers in Energy Research* 10:793776)** on CoverCress for sustainable aviation fuel. **All of this literature treats the cover crops as uniform whole-field establishments — there is no published spatial heterogeneity analysis of camelina/pennycress yield or any within-field placement framework.** Umn + 5

DSSAT-for-cover-crops literature is small but growing

Salmerón et al. (2014, *Agron. J.* 106:1283) simulated DSSAT cover crop–maize rotations under Mediterranean irrigation, reporting 31% N leaching reduction. **Birru et al. (2023, *Agron. J.* 115:1114)** and **Kharel et al. (2022, *Agric. Water Mgmt.*; 2025)** have extended DSSAT to cereal rye in the Midwest, with Kharel et al. (2025) predicting a 57% nitrate-loss reduction. Critically, **Kharel et al. (2022) explicitly wrote that DSSAT had not previously been used to simulate cereal rye as a cover crop without integration with other models** — a direct quotable gap statement that the proposal can build on. Wiley Online Library + 5

Spatially variable cover crop placement is essentially unstudied

This is the cleanest novelty claim. **Maldaner et al. (2020, *Comput. Electron. Agric.*)** used PlanetScope-derived rye/oat biomass as a proxy *to delineate zones for the subsequent cash crop* — closest in spirit, but inverse logic. Industry tools (EOSDA, GeoPard, John Deere Operations Center) sell variable-rate cover-crop prescriptions, but **peer-reviewed scientific evaluation of variable-rate cover crop seeding is essentially nonexistent**. No publication combines crop-model-driven zones + variable-rate cover crop seeding + bioenergy feedstock optimization in the US Midwest. ScienceDirect

Recommended publishable framing

The strongest framing positions the work as a **methodological and conceptual bridge** between the Brandes subfield-economics framework, the Basso/Thorp DSSAT precision-ag tradition, and the Forever Green/CoverCress biofuel cover-crop agenda. A target-tier draft title: **"Within-field heterogeneity as a pathway to scale biofuel cover crops in the US Corn Belt: crop-model-driven prescription mapping for variable-rate placement of cereal rye, winter camelina, and pennycress."** The key rhetorical contrast is with Brandes (2018): rather than converting 37% of cropland to perennials, **place winter annual biofuel cover crops selectively on zones identified by DSSAT simulation, preserving 100% of grain acreage**. Target journals in order of stretch: *Environmental Research Letters*, *GCB Bioenergy*, *Nature Food*, *Agriculture, Ecosystems & Environment*, then *Field Crops Research*, *Agricultural Systems*, or *Precision Agriculture* for more methods-focused framing.

Gaps that must be closed for rigor

A successful manuscript must address seven gaps explicitly. **First**, develop and publish CROPGRO-Camelina and (if attempted) CROPGRO-Pennycress cultivar coefficients via adaptation of CROPGRO-canola, calibrated against Forever Green/USDA-ARS Morris/IPREFER trial data — this is publishable as a separate methods paper and is non-negotiable. **Second**, replace or augment raw SSURGO with POLARIS plus a DSM/LiDAR layer, citing the 2025 *Agronomy Journal* result that DSM clustering reduces within-zone variability by 78–236%. **Third**, validate within-field cover-crop yield variability against remote-sensing-derived biomass (PlanetScope/Sentinel-2 calibrated to UAS measurements per Maldaner 2020) because cover-crop yield monitor data essentially does not exist. **Fourth**, demonstrate operational executability with the Montag/Hagie/Hiniker/John Deere VR equipment realistically deployable in the Midwest. **Fifth**, integrate explicit economics (CoverCress contract pricing, 45Z SAF credits, climate-smart commodity payments, cash-crop yield-drag risk) and a GREET-style LCA at subfield resolution — without this the biofuel framing is incomplete. **Sixth**, compare results explicitly against alternative strategies (whole-field cover crop, no cover crop, Brandes-style perennial subfield strips, STRIPS prairie strips) — reviewers will ask "why not perennials?" and the answer needs to be quantitative. **Seventh**, run multi-decade DSSAT to identify temporally stable zones rather than single-year artifacts, following Maestrini & Basso (2018, 2021); and acknowledge openly that APSIM and RZWQM2 are the historically dominant Midwest cover-crop platforms, justifying DSSAT as a deliberate choice tied to its CENTURY-SOM module, Sequence tool, and CROPGRO-canola starting point.

Conclusion: a defensible, novel project requiring substantial calibration investment

The project's core idea — using crop-model-driven within-field zones to prescribe spatially variable biofuel cover crop placement in the Midwest corn-soy system — is **genuinely novel at the intersection** and publishable in high-impact venues. Methodologically, it is defensible if SSURGO is supplemented with POLARIS and topographic data, Daymet-V4 weather is used, and the CENTURY-SOM module is properly initialized. Implementation feasibility is real for cereal rye and the cash crops, marginal for variable-rate seeding economics, and most importantly **gated on de novo DSSAT calibration of winter camelina and pennycress** — a substantial but publishable undertaking in itself. The strongest publishable framing positions the work as the winter-annual analogue to the Brandes/Bonner perennial subfield-diversification framework, exploiting the fact that biofuel cover crops can deliver feedstock without sacrificing grain area. The largest risks are (a) underestimating the camelina/pennycress modeling effort, (b) treating SSURGO as if it already captured within-field variability, and (c) failing to anchor the economic case in concrete biofuel feedstock contracts and SAF tax credits. Closed appropriately, this project is a credible candidate for *Environmental Research Letters* or *GCB Bioenergy*. FW Africa